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In this assignment, I explored real-world applications of face and object detection
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 5 Uses of Face/Object Detection
Face Unlock in Smartphones
Used to unlock devices securely by recognizing the user’s face.
Surveillance and Security Systems
Detects suspicious activities or unknown individuals in public places.
Attendance System
Automatically marks attendance using face recognition in schools or offices.
Autonomous Vehicles
Detects objects like pedestrians, vehicles, and traffic signals for safe driving.
Retail and Shopping Analytics
Tracks customer behavior, footfall, and product interaction in stores.
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🖥️ Designed Solution
🔹 Solution Name: Smart Face-Based Attendance System
🧠 Problem

Manual attendance systems are time-consuming and prone to errors or proxy attendanc

💡 Solution

Develop a system that uses face detection and recognition to automatically mark att

⚙️ How It Works
Capture real-time video using a camera
Detect faces using computer vision techniques
Match detected faces with stored student/employee data
Automatically mark attendance in a database
🛠️ Technologies Used
Python
OpenCV
Face Recognition library
Database (Excel / SQL)
✅ Benefits
Saves time
Prevents proxy attendance
Improves accuracy
Easy to manage records
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